

Representation and Communication of Digital Information II

Laboratórios de Sistemas e Serviços

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Exercises

Exercise 1: Markdown Basics

Create a file named `report.md` that includes the following elements: 1. A main title and two sub-sections. 2. A list of “Tasks for today” with at least 3 items. 3. A block of Python code that prints “Hello, Pandoc!”. 4. A table with two columns: Feature and Benefit. 5. A link to the University of Aveiro website.

Exercise 2: LaTeX Mathematics

Create a small LaTeX file named `math.tex` (or use an online editor like Overleaf) to typeset the following formulas: 1. The quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ 2. The definition of a derivative: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ 3. Maxwell’s Equations (just one of them, e.g., Gauss’s Law): $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$

Exercise 3: Using Pandoc for Conversion

In this exercise, you will use Pandoc to convert documents between formats. (Ensure pandoc and a LaTeX engine like `lualatex` or `pdftex` are installed).

1. Convert your `report.md` from Exercise 1 into an HTML file:

```
$ pandoc report.md -o report.html
```

2. Convert the same `report.md` into a PDF file:

```
$ pandoc report.md -o report.pdf
```

3. Convert the `report.md` into a LaTeX source file:

```
$ pandoc report.md -o report.tex
```

Exercise 4: Python Docstrings

1. Create a Python script named `math_tools.py`.
 2. Define a function `fibonacci(n)` that calculates the n-th Fibonacci number.
 3. Add a high-quality “Google-style” or “Numpy-style” docstring to the function, including:
 - A summary of what it does.
 - Descriptions of the arguments.
 - Description of the return value.
 - An example of usage.
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Exercise 5: Document Automation (Challenge)

Create a Makefile in a new directory that: 1. Finds all .md files in the directory. 2. Has a rule to convert each .md file into a .pdf using pandoc. 3. Has a clean rule to remove all generated PDFs. 4. Test it by running make and then make clean.